

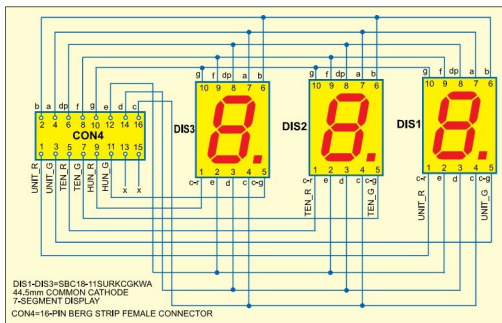


Fig. 2: Display section

process of the count-down timer is explained below. At power-'on', the software program starts looking for the red signal. As soon as the red signal is 'on', it starts counting the time in seconds until the signal goes off. At that moment, the count in seconds is saved as 'learn time.'

The software program again keeps waiting for the red signal. As soon as the red signal is sensed second time, it starts counting time again until the red light goes off. At that moment, the count in seconds is saved as 'verify time.' Now if 'learn time' and 'verify time' are same, this is treated as 'on time.'

Next time, as soon as the red signal is sensed, the software program knows that the red signal will remain 'on' for 'on time' based on the experience of past two readings. The count-down starts during this 'on' time.

The same process is applicable to the green traffic signal as well.

The working of the count-down timer is detailed in the next section.

Circuit description

Fig. 1 shows the circuit of the AVR-based traffic light count-down timer with dual-colour display. It consists of

the microcontroller, voltage regulator, optocoupler (PC817), red and green light input section, real-time clock and display driver section.

Microcontroller section. This section is built around ATMEGA ATmega328 microcontroller, which is a 28-pin DIP IC. The microcontroller IC can be programmed with embedded software using a standard programmer. A 16MHz oscillator is used to supply a timing reference. Out of 28 pins, only 20 are available for input/output (I/O).

We have used following I/O lines for different purposes:

1. Grids—six lines
2. Red/green input—two lines
3. Interface to the segment driver—five lines
4. PC interface—two lines

PD7 is connected to ground through a shorting jumper at CON1.

Voltage regulator section. This is a standard circuit which converts the 15V-35V DC input into 5V. Minimum and maximum DC inputs to the circuit are 15V and 35V, respectively. The regulated 5V DC provided by IC 7805 is used to operate the microcontroller and other digital ICs.

Optocoupler section. Two PC817 optocouplers are used to isolate the

input and output circuits. Red and green signal inputs are external signals coming from the traffic light control circuitry.

Red and green light inputs section. The input signals for the red and green lights varying from 15V to 30V are fed to the timer control circuit through diode-optocoupler combinations D1-IC3 and D2-IC4, respectively. When the red or green traffic light is 'on', the optocoupler conducts and the corresponding pin of the MCU goes low. When the red or green light is 'off', the corresponding pin of the MCU is high.

RTC section.

PCF8563 used in the RTC section is an I²C interface that uses SCL and SDA pins. Both these pins are available on the microcontroller too. RTC needs a Li-ion battery backup so that time is maintained even after power failure. The battery may need replacement once in three years.

CLKOUT signal of IC PCF8563 drives BC847. As a result, LED4, connected to the collector of transistor BC847, flashes at a frequency of 1 Hz to indicate proper working of RTC IC PCF8563. Note that the RTC section was not used when testing this project at EFY Lab.

Display driver. We have used three dual-colour, common-anode, seven-segment LED displays. Each display has eight segment pins and two grid points (C- and C-g). For three-digit display, we have used eight segments and six grid outputs. As the 15V and up to 100mA current required to drive the display cannot be supplied by the microcontroller, we have used a segment driver and a grid driver (TD62783). The six outputs of TD62783 are used to drive the six grids (three for red colour and three for green colour).

TLC5916 is used as the seven-